

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <fcntl.h>
4  #include <sys/mman.h>
5  #include <unistd.h>
6  #include <semaphore.h>
7  #include <errno.h>
8
9  #define SHM_NAME "/arche_noah_shm"
10 #define SEM_MUTEX "/arche_noah_mutex"
11 #define SEM_FULL "/arche_noah_full"
12 #define N 10
13
14 struct shared_data
15 {
16     int bookings[N];
17     int count;
18 };
19
20 void sem_wait_intr(sem_t *sem)
21 {
22     while (sem_wait(sem) == -1)
23     {
24         if (errno != EINTR)
25         {
26             perror("sem_wait");
27             exit(1);
28         }
29     }
30 }
31
32 int main()
33 {
34     int shm_fd = shm_open(SHM_NAME, O_CREAT | O_RDWR, 0666);
35     if (shm_fd == -1)
36     {
37         perror("shm_open");
38         exit(1);
39     }
40
41     ftruncate(shm_fd, sizeof(struct shared_data));
42
43     struct shared_data *data = mmap(NULL, sizeof(*data),
44                                     PROT_READ | PROT_WRITE,
45                                     MAP_SHARED, shm_fd, 0);
46
47     sem_t *mutex = sem_open(SEM_MUTEX, O_CREAT, 0666, 1);
48     sem_t *full = sem_open(SEM_FULL, O_CREAT, 0666, 0);
49
50     if (mutex == SEM_FAILED || full == SEM_FAILED)
51     {
52         perror("sem_open");
53         exit(1);
54     }
55
56     printf("Master waits for complete Buchungsliste...\n");
57
58     sem_wait_intr(full);
59
60     sem_wait_intr(mutex);
61     printf("Buchungsliste complete:\n");
62     for (int i = 0; i < N; i++)
63     {
64         printf("Space %d: Client %d\n", i, data->bookings[i]);
65     }
66     sem_post(mutex);
67

```

alles schliessen nicht vergessen, ansonsten signal richtig behandelt

alles schliessen nicht vergessen (-1)

```
68 | munmap(data, sizeof(*data));
69 | close(shm_fd);
70 |
71 | shm_unlink(SHM_NAME);
72 | sem_unlink(SEM_MUTEX);
73 | sem_unlink(SEM_FULL);
74 |
75 | return 0;
76 | }
77 |
```